

A Weak Law for Random Semigroup Products on Every Separable Banach Space

Candidate full solution to Conjecture 2.5 of arXiv:2410.07417

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Status. Candidate full solution, likely valid pending expert review. Under the standard random-element meaning of i.i.d., the proof gives uniform-in-time L^1 , hence in-probability, strong operator convergence on every separable Banach space.

1 Source problem

Dzhenzher and Sakbaev study weakly measurable random generators on ℓ_p , $1 \leq p \leq 2$, whose operator norms are uniformly bounded. Their Theorem 2.4 proves convergence in the strong topology of ℓ_2 . On page 4 they formulate the following conjecture [1]:

Under the assumptions of Theorem 2.4, $e^{A_1 t/n} \dots e^{A_n t/n}$ converges in probability in SOT of ℓ_p to $e^{(\mathbb{E}A)t}$ uniformly for t in any compact interval.

Their precise meaning is that for every $x \in \ell_p$, $T > 0$, and $\varepsilon > 0$,

$$\mathbb{P} \left\{ \sup_{0 \leq t \leq T} \left\| \left(e^{A_1 t/n} \dots e^{A_n t/n} - e^{(\mathbb{E}A)t} \right) x \right\|_p > \varepsilon \right\} \longrightarrow 0.$$

The relevant source passage is reproduced below.

e^{At} for $t \geq 0$. A sequence $W_n(t)$ of compositions of random semigroups generated by weakly measurable operators $\Omega \rightarrow \mathcal{L}(\ell_p)$ converges in probability in SOT of ℓ_q to a random semigroup $W(t)$ uniformly for $t > 0$ in any segment if for any $x \in \ell_p$, $T > 0$ and $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P} \left\{ \sup_{t \in [0, T]} \|(W_n(t) - W(t))x\|_q > \varepsilon \right\} = 0.$$

The following theorem is known in the case of unitary semigroups generated by self-adjoint operators. We consider the case of bounded generators.

Theorem 2.4 (proved in §3). *Let $1 \leq p \leq 2$. Let $A: \Omega \rightarrow \mathcal{L}(\ell_p)$ be a generator of a semigroup e^{At} such that $\|A\|_p$ lies in a bounded ball. Let $A_1, A_2, \dots$ be a sequence of i.i.d. generators that distributions coincide with the distribution of A . Then $e^{A_1 t/n} \dots e^{A_n t/n}$ converges in probability in SOT of ℓ_2 to $e^{\mathbb{E}A t}$ uniformly for t in any segment.*

It is interesting to obtain the LLN in SOT of ℓ_p not ℓ_2 . This leads us to the following conjecture.

Conjecture 2.5. *Under the assumptions of Theorem 2.4, $e^{A_1 t/n} \dots e^{A_n t/n}$ converges in probability in SOT of ℓ_p to $e^{\mathbb{E}A t}$ uniformly for t in any segment.*

Some steps for proof of this conjecture are made in §4, see e.g. Theorem 4.1.

2 Proof intuition

At scale t/n , replacing each exponential by $I + tA_i/n$ costs only $O(n^{-2})$ per factor and therefore $O(n^{-1})$ for the whole product. The resulting Euler product has an exact expansion by ordered subsets:

$$\prod_{i=1}^n \left(I + \frac{t}{n} A_i \right) x = \sum_{k=0}^n t^k \frac{1}{n^k} \sum_{i_1 < \dots < i_k} A_{i_1} \cdots A_{i_k} x.$$

For each fixed k , the coefficient is a nonsymmetric Banach-valued U -statistic. A direct simple-function argument shows that it converges in L^1 to $(\mathbb{E}A)^k x / k!$ on *any* separable Banach space: disjoint index tuples are independent, and only $O(n^{2k-1})$ pairs overlap. Finally, boundedness supplies the uniform factorial majorant $\rho^k \|x\| / k!$. Thus fixed-order convergence can be summed over every k , uniformly for t in a compact interval. This bypasses all type, smoothness, and Hilbert-space variance estimates.

3 An ordered U -statistic lemma

Lemma 1. *Let S be a measurable space, let $\xi_1, \xi_2, \dots$ be i.i.d. S -valued random elements with law μ , and let X be a separable Banach space. If $h : S^k \rightarrow X$ is Bochner integrable, then*

$$\frac{1}{n^k} \sum_{1 \leq i_1 < \dots < i_k \leq n} h(\xi_{i_1}, \dots, \xi_{i_k}) \xrightarrow[n \rightarrow \infty]{L^1(X)} \frac{1}{k!} \int_{S^k} h d\mu^k. \quad (1)$$

No symmetry of h is required.

Proof. First take $h = \mathbf{1}_D$ for a measurable $D \subseteq S^k$. The expectation of the left side is $n^{-k} \binom{n}{k} \mu^k(D)$, which tends to $\mu^k(D) / k!$. Two summands whose index sets are disjoint are independent. Among the $\binom{n}{k}^2 = O(n^{2k})$ ordered pairs of increasing k -tuples, only $O_k(n^{2k-1})$ pairs have an intersecting index set. Since every covariance has absolute value at most one, the variance of the normalized sum is $O_k(n^{-1})$. Hence (1) holds in L^2 , and therefore in L^1 , for indicator kernels. By linearity and the triangle inequality it holds for finite simple X -valued kernels.

For general h , choose finite simple g with $\int \|h - g\| d\mu^k < \delta$. If $U_n(h)$ denotes the left side of (1), then

$$\mathbb{E} \|U_n(h) - U_n(g)\| \leq \frac{\binom{n}{k}}{n^k} \int \|h - g\| d\mu^k \leq \frac{\delta}{k!}.$$

The distance between the two proposed limits is also at most $\delta / k!$. Let first $n \rightarrow \infty$ and then $\delta \downarrow 0$. $\square$

4 Full theorem

We use the following standard measurable formulation. On the norm ball

$$\mathcal{B}_\rho = \{T \in \mathcal{L}(X) : \|T\| \leq \rho\}$$

put the strong operator topology and its Borel σ -field. When X is separable this topology is metrizable on $\mathcal{B}_\rho$ by evaluating on a countable dense subset of X .

Theorem 2 (all-separable-space weak law). *Let X be a separable real or complex Banach space. Let $A, A_1, A_2, \dots$ be i.i.d. $\mathcal{B}_\rho$ -valued random elements for the strong-operator Borel σ -field. Define $B \in \mathcal{L}(X)$ by*

$$Bx = \mathbb{E}(Ax), \quad x \in X.$$

Then for every $x \in X$ and $T > 0$,

$$\mathbb{E} \sup_{0 \leq t \leq T} \left\| e^{A_1 t/n} \dots e^{A_n t/n} x - e^{Bt} x \right\| \longrightarrow 0. \quad (2)$$

Consequently the convergence is uniform on $[0, T]$ in probability in SOT.

Proof. Fix x and T . Write $h = t/n$. Taylor's formula and $\|A_i\| \leq \rho$ give

$$\|e^{hA_i} - (I + hA_i)\| \leq \frac{\rho^2 h^2}{2} e^{\rho h}, \quad \|e^{hA_i}\|, \|I + hA_i\| \leq e^{\rho h}.$$

A telescoping product estimate therefore yields, deterministically and uniformly for $0 \leq t \leq T$,

$$\left\| \prod_{i=1}^n e^{tA_i/n} x - \prod_{i=1}^n \left(I + \frac{t}{n} A_i \right) x \right\| \leq \frac{\rho^2 T^2 e^{\rho T}}{2n} \|x\|. \quad (3)$$

Expand the Euler product without commuting its factors:

$$\prod_{i=1}^n \left(I + \frac{t}{n} A_i \right) x = \sum_{k=0}^n t^k C_{n,k}(x), \quad C_{n,k}(x) = \frac{1}{n^k} \sum_{i_1 < \dots < i_k} A_{i_1} \dots A_{i_k} x. \quad (4)$$

For fixed k , apply Lemma 1 to $h_k(T_1, \dots, T_k) = T_1 \dots T_k x$. This kernel is strongly measurable: on $\mathcal{B}_\rho^k$, multiplication followed by evaluation at x is continuous for the product SOT, because all operators have the common norm bound ρ . It is bounded by $\rho^k \|x\|$. Iterated Bochner integration gives

$$\int h_k d\mu^k = B^k x,$$

so

$$C_{n,k}(x) \xrightarrow{L^1(X)} \frac{1}{k!} B^k x. \quad (5)$$

Moreover, for every outcome,

$$\|C_{n,k}(x)\| \leq \frac{\binom{n}{k}}{n^k} \rho^k \|x\| \leq \frac{\rho^k \|x\|}{k!}, \quad \left\| \frac{B^k x}{k!} \right\| \leq \frac{\rho^k \|x\|}{k!}. \quad (6)$$

For an integer K , equations (5)–(6) imply

$$\begin{aligned} \mathbb{E} \sup_{0 \leq t \leq T} \left\| \sum_{k=0}^n t^k C_{n,k}(x) - e^{tB} x \right\| \\ \leq \sum_{k=0}^K T^k \mathbb{E} \left\| C_{n,k}(x) - \frac{B^k x}{k!} \right\| + 2\|x\| \sum_{k>K} \frac{(\rho T)^k}{k!}. \end{aligned}$$

For fixed K the first term tends to zero; the second tends to zero as $K \rightarrow \infty$, uniformly in n . This proves the Euler-product version of (2). Combining it with (3) proves (2). Markov's inequality gives the asserted convergence in probability. The displayed supremum is measurable because the paths are continuous and the supremum may be taken over rational $t \in [0, T]$. $\square$

Corollary 3 (source conjecture). *Let $1 \leq p < \infty$. If the weakly measurable, uniformly bounded operators in Conjecture 2.5 are i.i.d. in the standard random-element sense for the ℓ_p strong-operator Borel σ -field, then the conjecture holds. In fact, its convergence is uniform-in-time L^1 as in (2).*

Proof. Because ℓ_p is separable, weak measurability of A implies strong measurability of every Ax by Pettis’s measurability theorem. On a fixed operator-norm ball, a countable dense set of vectors metrizes SOT; hence A is a Borel random element for that topology. The Pettis mean used in [1] agrees pointwise with the Bochner integrals $\mathbb{E}(Ax)$. Apply Theorem 2. $\square$

5 Conventions, limitations, and verification

Independence convention. The proof uses ordinary independence of the operator random elements, or equivalently independence of the σ -fields generated by all SOT evaluations. This is the standard meaning of “i.i.d. random generators” and is the meaning adopted explicitly in the authors’ later paper [2]. The 2024 source also gives a bespoke sentence defining independence by requiring one scalar rank-one observation $\langle f_i, A_i x_i \rangle$ from each operator to be independent. Taken literally, that sentence does not visibly imply independence of the full evaluation σ -fields, because the class of rank-one observations is not closed under finite joint cylinders. Thus a reviewer should either confirm that the sentence was intended as shorthand for standard random-element independence, or regard Corollary 3 as a precise correction of that convention. No geometric assumption on X is hidden in the proof.

Computational check. The accompanying script tests noncommuting finite-dimensional examples. It compares exact exponential products, Euler products, and $e^{t\mathbb{E}A}$ on a time grid, and separately checks the first three ordered coefficients. Observed mean errors decrease with n . This is a sanity check only; the proof above is analytic and dimension-free.

6 Literature and novelty check

A bounded search completed on 17 August 2026 used the run’s registry, solution, attempt, and proof-gap indexes; the local arXiv corpus; exact-title and author queries for arXiv:2410.07417; and close variants of “random semigroup weak law”, “Banach-valued ordered U-statistic”, and “strong operator law of large numbers”. The principal later match is the same authors’ 2025 preprint [2]. It proves almost-sure SOT convergence on uniformly smooth Banach spaces (hence ℓ_p , $1 < p < \infty$), proves an all-Banach WOT theorem, and states an ℓ_1 consequence via Schur’s theorem. Its random operators, however, are Bochner measurable as elements of the *operator-norm* Banach space. That is a stricter measurability hypothesis than the weak/SOT-measurable hypothesis of [1] and of Corollary 3. No source was located that states the all-separable-space uniform L^1 theorem above or proves the original weakly measurable formulation by this ordered- U -statistic argument. Novelty confidence is moderate pending a specialist search.

7 Human review recommendation

The decisive points to audit are: (i) the L^1 simple-kernel approximation in Lemma 1; (ii) continuity and measurability of the ordered composition kernel on a common SOT norm ball; and (iii) whether the source’s use of “i.i.d.” is intended in the standard random-element sense. The Euler estimate

and factorial-tail resummation are deterministic. If (iii) is accepted, the packet is a full solution of Conjecture 2.5 and a strict space-class strengthening.

References

- [1] S. Dzhenzher and V. Sakbaev, *Quantum law of large numbers for Banach spaces*, arXiv:2410.07417 (2024), Conjecture 2.5 on p. 4.
- [2] S. Dzhenzher and V. Sakbaev, *The Strong Law of Large Numbers for random semigroups on uniformly smooth Banach spaces*, arXiv:2507.07658 (2025), Theorems 2.1 and 2.2.